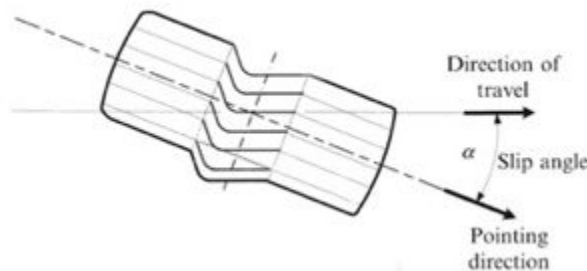
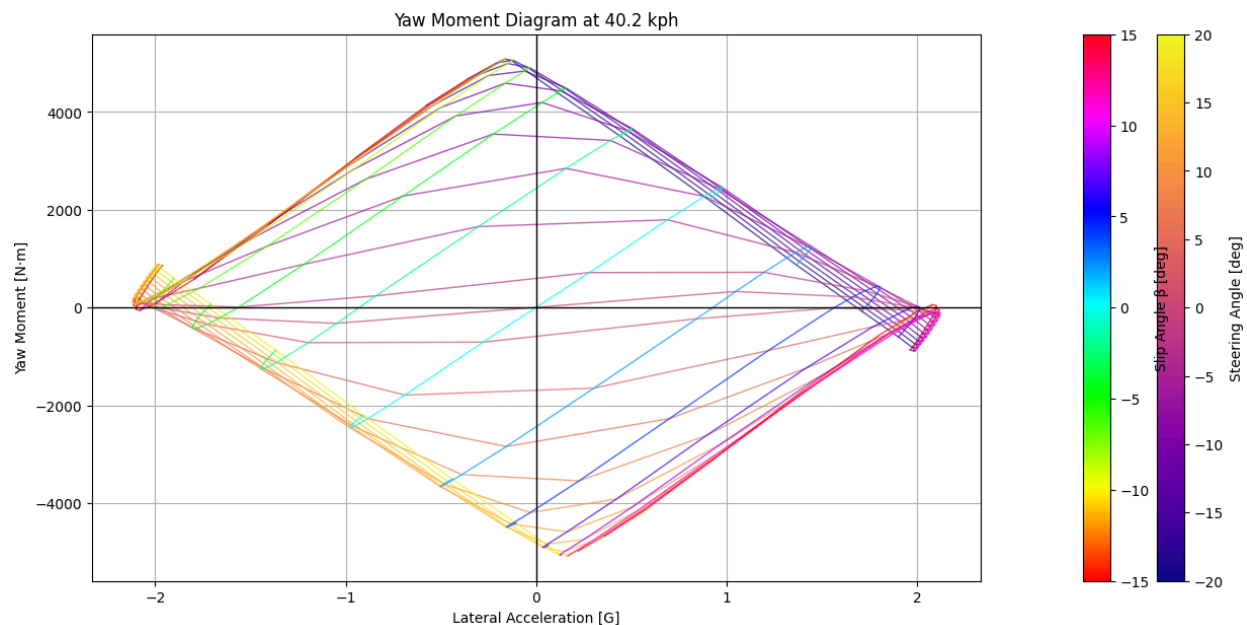


A YMD (Yaw Moment Diagram) is a diagram generated from plotting points of selected tyre slip and chassis slip angles.

- What is a chassis slip angle?
 - When racing the chassis of the car is rarely headed EXACTLY where it is pointed, it is usually in some kind of drift, the angle of the drift (the angle between where the car is pointed vs headed) is the chassis slip angle
- What is a tyre slip angle?
 - Tyres generate lateral grip by being misaligned from the road and deforming the rubber, the angle between where the tyre is pointed and where the tyre is headed is the slip angle of the tyre.



The YMD is plotted onto a Yaw Moment vs Lateral Acceleration graph. There are only a few key KPIs that we care about but here's a brief description of how to read the graph as a whole. The Y-axis represents how much more we can turn the car if we are already at a given lateral acceleration.



What this YMD is showing is that when we are at 1G of lateral acceleration, we can induce another ~2000 Nm of yaw moment to the car.

- What is yaw moment?
 - A moment is a force applied over a distance, the yaw moment is the force of the tyres (and technically aero but lets ignore that for now) applied at the distance

from the center of gravity/mass. We call this distance a for front axle to C_g and b for rear axle to C_g which makes the yaw moment about Front Axle $F_y * a$ and similar for the rear. (THIS IS AN OVERSIMPLIFICATION)

Now onto the KPIs that we care about. Since we will need our front and rear load transfer sensitivity (more on this in the tutorial) we can log TLLTD and treat it as a diagnostic KPI since its simple to calculate independently. The KPIs that are unique to the YMD are Limit Balance, Steady State Grip Limit, Control, and Stability.

The Steady State Grip Limit (Grip Limit) is the maximum lateral acceleration that the car achieves on the YMD when the Yaw Moment is 0 i.e. under balanced conditions. Measured in G's it is a representation of the cars theoretical performance.

- What does steady state mean?
 - There are two types of states the car can be in, steady state and transient.
 - Transient situations are when the car is actively changing its movement in some way, this is on acceleration, corner entry/exit, any time that the driver is changing the steering angle, throttle, or brake. This is Autocross and what most people think of as normal driving situations.
 - Steady State situations are after the car has settled into its new state. So driving in a straight line at a constant velocity is steady state, cornering at a constant velocity is steady state (Skidpad). There are no new forces acting from or on the car to change its state of being. It is like finding a local minimum in a graph.

The Limit Balance is what tells us how over/understeery the car is under the given conditions of the YMD. We tend to look at the positive side of the YMD for some reason so everything will be in reference to that (the right side) of the YMD. The limit balance is the offset from the X-axis (0 yaw moment) at the point of maximum lateral acceleration. So referring to the previous figure, since the point of maximum lateral acceleration is below the x-axis that tells us that the car at that speed with that setup is understeery.

- What is understeer/oversteer?
 - Not answering that here, it's definitely somewhere else. See also: That one top gear clip.

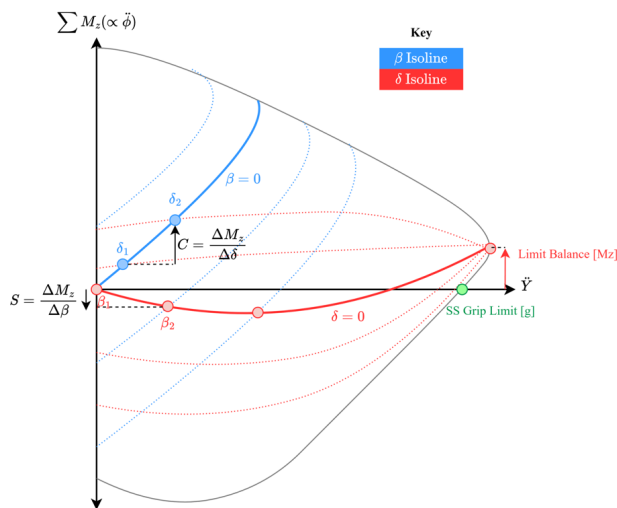
So now is the time we have to acknowledge the lines in the graph. The lines are plotted along lines of constant chassis slip angle and constant tyre slip angle. The flat(ish) lines through the center are constant steering angles while the lines that are closer to 45 degrees through the center are constant chassis slip angles.

Control is the slope of the chassis slip angle lines where stability is the slope of the constant steering angle line. For both of these we force the solving of the 0 iso line for their respective regimes (0 steering angle and 0 chassis slip angle). And use the points right around the origin to gather the values.

Control is defined in Nm/deg (deg of steering) and represents how much authority changing the steering angle has on the direction of the car. We define control as being relative to the angle of

steering at the contact patch of the tyre, so it is really relative to slip angle not steering angle (at the steering wheel). This has gotten us into trouble with Claude but his website has a link to a YMD doc that has control defined this way so we are defensible against him. Stability is defined in Nm/deg (deg of chassis slip) and represents how easily the car will return to its baseline when excited by an external factor, i.e. a gust of wind.

KPI AND PERFORMANCE METRICS



Steady State Grip Limit	Limit Balance
- Maximum lateral acceleration where $\sum M_z = 0$ - The performance limit in a 'balanced case' - Acceleration [g]	- Moment at the maximum lateral acceleration - Indicator of vehicle balance - Moment [N.m]
'Control'	'Stability'
- Slope of M_z vs δ for a given isoline of constant body slip angle - Indicator of moment generated for a given steering input. - Steady state 'control' KPI [N.m/deg]	- Slope of M_z vs β for a given isoline of constant steering angle - Indicator of moment generated for a given body slip angle that returns the vehicle towards equilibrium [N.m/deg]

